Sahaj Bhadja

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Summary

Early-career Software Engineer graduating May 2026 with 2+ years shipping full-stack and backend systems, plus hands-on Solidity, DeFi, smart-contract testing, TypeScript, React, distributed systems, and peer-reviewed blockchain security research.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected multi-tenant data infrastructure on Azure SQL and Blob Storage with row-level isolation, supporting 500+ users while reducing indexed-query latency from 1.2 seconds to 350 milliseconds reliably.
- Restructured fault-tolerant asynchronous orchestration with explicit state transitions and failure isolation, processing 1K+ jobs while tripling throughput and preventing cascading failures across dependent production services under load.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent backend workflows with secure file validation, processing 1K+ files per batch through AWS RDS and REST APIs with isolated, observable, and recoverable execution reliably at scale.
- Implemented production services handling 10K+ daily requests for 150+ users, applying CI/CD-backed testing and deployment workflows to reduce turnaround 60% while maintaining stable peak-load application performance in production.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a Python simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% through validated data transformations across controlled scenarios and repeatable system-behavior benchmark tests in Python.
- Developed a spatial-temporal encoder combining map geometry and trajectory context, improving closed-loop stability 15% through regression testing, systematic failure analysis, documented evaluation, and root-cause debugging in production.

Aasma Technology Solutions **August 2023 – July 2024**
Software Development Engineer, SecureSync *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments consistently.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% through reusable services, automated validation, and structured code reviews for production releases.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellow *India*

- Co-authored two peer-reviewed IEEE papers on decentralized user authentication and post-quantum blockchain security using smart contracts, FALCON signatures, and consensus design, then presented findings at IISc Bengaluru.
- Benchmarked a FALCON-DPoSB framework across 100–1,000 transaction workloads, sustaining 957.8 TPS at 224 ms latency while preserving transparent witness selection against a classical ECDSA baseline at scale.

Skills

Languages: Solidity, TypeScript, JavaScript, Python, Java, C++, C#, SQL, HTML, CSS, Bash
Web3 & Backend: DeFi, Smart Contracts, AMMs, EVM, FALCON, Consensus Algorithms, REST APIs, Microservices, Distributed Systems
Frontend & Data: React, Next.js, Node.js, PostgreSQL, SQL Server, MySQL, MongoDB, WebSockets, Data Modeling
Infrastructure & Quality: AWS, Azure, Docker, Kubernetes, Git, CI/CD, Unit Testing, Integration Testing, Debugging, Code Reviews

Projects

SwapGuard | *Solidity, EVM, AMM, Ethers.js, Ganache*

- Built, compiled, and deployed a Solidity AMM safeguard contract to a local EVM, validating constant-product quotes, configurable fee bounds, owner authorization, deadlines, and minimum-output slippage protection across repeatable tests.

Quantum-Resistant Blockchain Technology Paper | *FALCON, DPoSB, Borda Count*

- Proposed a post-quantum blockchain framework combining FALCON signatures with behavior-aware DPoSB consensus and Borda Count witness selection; evaluated throughput and latency against ECDSA while preserving decentralized, transparent validator governance.

Blockchain for User Authentication Paper | *Smart Contracts, DLT, IoT, RBAC*

- Analyzed decentralized, smart-contract-driven identity verification for users and IoT devices, including immutable records, self-sovereign identity, and role-based access control without centralized third-party identity providers or password databases.